

Table IV.4: Statistical Measure and Justification

Statistical Measure	Applied In	Justification
Median	All scenarios	Robust to outliers; reflects typical performance under right-skewed timing distributions
Interquartile Range (IQR)	All scenarios	Describes spread of central 50% of observations without distortion from extreme values
5th–95th Percentile Range	All scenarios	Captures broader variability and worst-case performance without distortion from outliers
Skewness	All scenarios	Quantifies asymmetry in timing distributions, expected to be positive due to system-level delays
Mean	Scenario 3 (AES)	Used alongside median where throughput averages are the primary reporting metric
Standard Deviation	Scenario 3 (AES)	Measures dispersion around the mean; retained where average-based reporting is appropriate
95% Confidence Interval	Scenario 3 (AES)	Provides a range of uncertainty around the estimated mean execution time
Throughput (MB/s)	Scenarios 2 & 3	Input size divided by execution time; describes sustained processing efficiency
Shapiro-Wilk Normality Test	Scenario 3 & 4 (AES and Hybrid)	Assesses whether timing distributions approach normality or exhibit significant skew. Applied to AES where throughput-based reporting requires understanding of distributional shape, and to the hybrid workflow where total and component-level latency distributions are tested to confirm whether median or mean-based interpretation is more appropriate
Component-level Timing	Scenario 4 (Hybrid)	Decomposes total latency into ECDH, ML-KEM, and KDF stages to identify the dominant cost driver